INCH-POUND

MIL-P-80186B
20 March 1991
SUPERSEDING
MIL-P-80186A
27 September 1985

MILITARY SPECIFICATION

POSITIONERS, WELDING, ROTATING TABLE, CANTILEVER ARM DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rotating table type welding positioners designed with the table off-set from the base to provide 360 degree table rotation.

1.2 Classification. This specification covers positioners of the following classes and sizes. The class and size to be furnished shall as specified (see 6.2).

Class A - Manual table rotation with manual tilt.

Class B - Powered table rotation with manual tilt.

Class C - Powered table rotation and powered tilt.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3436
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

Table sizes and capacities.

Class A

Size 1 - 50 pounds capacity, bench mounted.

Size 1a- 100 pounds capacity, bench mounted

Size 2 - 200 pounds capacity, bench mounted.

Size 3 - 300 pounds capacity, roller base mounted.

Size 4 - 500 pounds capacity, floor mounted.

Size 5 - 1000 pounds capacity, floor mounted.

Size 6 - 2000 pounds capacity, floor mounted.

Size 7 - 4000 pounds capacity, floor mounted.

Class B

Size 1 - 50 pounds capacity, bench mounted.

Size 1a- 100 pounds capacity, bench mounted.

Size 2 - 200 pounds capacity, bench mounted.

Size 3 - 300 pounds capacity, roller base mounted.

Size 4 - 500 pounds capacity, floor mounted.

Size 5 - 1000 pounds capacity, floor mounted.

Size 6 - 2000 pounds capacity, floor mounted.

Size 6a- 3000 Pounds capacity, floor mounted.

Size 7 - 4000 pounds capacity, floor mounted.

Class C

Size 5 - 1000 pounds capacity, floor mounted.

Size 6 - 2000 pounds capacity, floor mounted.

Size 6a- 3000 Pounds capacity, floor mounted.

Size 7 - 4000 pounds capacity, floor mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Source: https://assist.dla.mil Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete positioner shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.2 Design. The positioner shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The positioner shall be capable of supporting, rotating, elevating, and tilting workpieces with accuracy and welding repeatability for effective welding operation. The positioner shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein, but are a part of the manufacturer's current model shall be included in the positioner furnished. The positioner shall be of the cantilever arm design, with the arm supported at the upper end of the vertical column. The worktable shall be located on the projecting end of the arm off-set 90 degrees from the column. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate rotation and tilting motion shall be graduated in increments not larger than one degree. Dials and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.2 Reclaimed-materials. The positioner may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the positioner. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The positioner and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and electrical operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the positioner, whether or not specifically required herein shall conform with all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers, guards, or other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial positioner supplied to the commercial market shall be provided for the point of operation and all other parts of the positioner that present

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the positioner (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the positioner. All guards shall allow access to the guarded part.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise level emitted by the positioner when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type II sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the positioner and its components, the positioner shall meet all applicable Environmental Protection Agency (EPA) and Occupational Safety and Health Standard (OSHA) restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the positioner at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Mercury restriction. The positioner shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the positioner.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or positioner.

3.3 Construction. The positioner shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the positioner to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the positioner and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The positioner shall consist of not less than a base and column assembly, frame and base assembly, table, a rotation and tilting system, self-locking worm and wormgear system, and all the necessary controls to comprise a complete positioner system.

3.4.1 Bench mounted base and column assembly. Class A, size 1, 1a, and 2 positioners shall be configured for bench mounting. Tilt action around the arm axis of the positioner shall be accomplished through the arm mounting supported with the work table positioned face up at "0" degrees tilt, and the horizontal plane of the table top shall be higher than any other portion of the positioner. The base and column assembly has provision for the arm to revolve 360 degrees around the column with means for locking in any position.

3.4.1.1 Roller supported base and column assembly. Class A and B, size 3 positioner shall be mounted on a "T" type base. The cross section of the base shall be approximately 35 inches in length and the supporting section approximately 30 inches long by 10 inches wide. The positioner column shall be mounted at the center of the cross section of the"T". The base shall be mounted on 3 inch diameter steel casters with 2 swivel asters located on the cross section and 2 rigid casters located on the Supporting section. When specified (see 6.2), foot operated locks shall be provided for each swivel caster. Tilt action, horizontal plane table top height, and arm rotation requirements for the size 3 positioner shall be the same as specified for sizes 1 through 2 positioners.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the currentversion before use.

3.4.1.2 Floor mounted base and column assembly. Classes A and B, sizes 4 through 7 positioners, and Class C, 5 through 7 positioners, shall be configured for floor mounting. The arm shall be hinged at two points to provide compensating angle adjustments of from "0" degrees to not less than 30 degrees. One hinge point shall be through the arm mounting support and one hinge point shall be between the arm mounting support and the worktable. The table tilt action shall be accomplished through a self-locking wormgear drive system on the portion of the arm mounted on the column. The hinge shall be such that the work table can be adjusted to the horizontal portion regardless of the degree of angle. Positioner tilt arm shall revolve 360 degrees around the horizontally base and 360 degrees vertically and shall have means for locking in any position.

3.4.2 Frame and base assembly. The frame and base assembly shall be a rigid steel weldment constructed to maintain alignment and to minimize deflection. The frame shall consist of a base, column, and arm assembly floor mounted positioners shall be capable of supporting maximum capacity loads without the use of counterweights or floor attachments. The base shall be provided with means for bolting the base to the floor or work bench. Class A and B, sizes 1, 1a, 2, and 7 positioners shall have either a fixed or adjustable column. Sizes 3, 4, 5, 6, and 6a of class A and B positioners and sizes 5, 6, 6a, and 7 of class C positioners shall be provided means for vertically adjustment of the table. Vertically adjustment shall be in addition to the arm angle adjustment. The table height shall be measured with the angle adjustment at "0" degrees and with the arm and table on a level plane. Each positioner shall be provided with an electrical grounding block on the base or column of the positioner suitable for bolting an electrical ground cable.

3.4.3 Table. The work table or face plate shall be of sufficient strength for performing circular downhand, irregular, and other welding operations at capacity load conditions. Unless otherwise specified (see 6.2), the table shall be circular in shape, and shall be not less than the dimensions specified in table I. When specified (see 6.2), the work surface shall be machined flat to within ± 1/32 of an inch. The table shall be mounted on a spindle. The spindle shall be supported on sealed anti-friction bearings. The class A positioners shall be provided with a hand operated adjustable friction brake system and an indexing pin locking device for locking the table in not less than eight equally spaced positions. The class B and C motorized positioner work table and motorized tilting work table shall be provided with a self-locking wormgear system to prevent vertical tilt action around the cantilever arm and horizontal table rotation. Positioners shall be provided with a dynamic braking system and shall be powered by a variable speed motor drive. Each positioner table shall be provided with either four radial T slots conforming to ANSI B5.1 or four equally spaced unthreaded bolt holes for mounting the workpiece.

Source: https://assist.dla.mil Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

Table I. Positioner sizes and capacities.

Class A manual table rotation and manual tilt.

Positioner size	Table capacity load (lbs)	Table diameter (in)	Table off-set center of column to center of table (in)	Table center of gravity (cg)	Table height (in)	Table rotation speed
1	50	5	10	4	10	manual
1a	100	6	12	12	10	--
2	200	6	20	12	12	--
3	300	6	20	12	12 to 31	--
4	500	6	32	12	22 to 31	--
5	1000	8 to 30	32	12	30 to 40	--
6	2000	8 to 36	35	12	33 to 39	--
7	4000	12 to 42	51	6x6	41	--

NOTE: All requirements shall be not less than the figures shown. Where a range is shown, the requirements shall range from not more than the minimum to not less than the maximum figures shown in the table.

3.4.3.1 Table drive. Class B and C table motorized drive systems shall be powered by a high frequency protected silicon control rectifier (SCR) solid-state electronic drive or variable speed SCR direct current (DC) motor drive. The system shall be totally enclosed with a self-locking wormgear or spur gear reducer. The motor shall be provided with dynamic braking and overload protection. The table rotation speed shall be reversible and infinitely variable over the entire range of the base speed specified in tables II and III. The speed control shall be accurate within 3 percent of the base speed. The drive system shall not restrict the tilt action, angle adjustment, or reduce the capacity of the positioner.

3.4.3.2 Controlled atmosphere operation table. The drive mechanism, controls, and lubrication system of the class A and B, size 1 positioner shall be completely enclosed and capable of operating in a controlled atmosphere chamber. Rotation accuracy shall be within 3 percent of the base speed. When specified (see 6.2), a remote control system shall be furnished.

3.4.3.3 Normal operating table drive. The powered rotation (rpm) range for class B and class C positioners shall conform to tables II and III. The positioner drive system shall be controlled by a remote foot operated control switch with forward, reverse, stop operations, and dynamic braking for reversing the electronic drive. The table shall be provided with a toggle switch to select between hand and foot control operation. Powered rotation shall also include rapid travel push button speed potentionmeter, tilt forward and reverse button, and pilot light. The remote foot operated switch shall be provided with not less than eight feet of cable wired through an on/off switch located on the positioner. The forward and reverse selector switch for the powered table rotation may be located on the positioner or in the electrical control cabinet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

TABLE II. Positioner sizes and capacities.

Class B powered table rotation and manual tilt.

Positioner size	Table capacity load (lbs)	Table diameter (in)	Table off-set center of column to center of table (in)	Table center of gravity (cg)	Table height (in)	Table rotation speed (RPM)	Table rotation torque (in/lbs)
1	50	5	10	4	10	0.2 to 10	50
1a	100	6	12	12	10	0.2 to 6	400
2	200	6	20	12	12	0.2 to 6	1500
3	300	6	20	12	22 to 31	0.2 to 6	1500
4	500	6	32	12	22 to 31	0.2 to 6	1500
5	1000	8 to 30	32	12	30 to 40	0.2 to 6	1500
6	2000	8 to 30	35	12	30 to 40	0.2 to 6	1500
6a	3000	10 to 42	49	6x6	12 to 42	0.2 to 6	2000
7	4000	12 to 42	51	6x6	41	0.2 to 1.3	2000

TYPE III. Positioner sizes and capacities.

Class C powered table rotation and powered tilt.

Positioner size	Table capacity load (lbs)	Table diameter (in)	Table off-set center of column to center of table (in)	Table center of gravity (cg)	Table height (in)	Table rotation speed (RPM)	Table rotation torque (in/lbs)	Table tilt torque (in/lbs)
5	1000	8 to 30	32	12	30 to 40	0.2 to 6	1500	6000
6	2000	8 to 36	35	12	33 to 39	0.2 to 6	1500	12000
6a	3000	0 to 42	49	6x6	12 to 42	0.2 to 6	2000	18000
7	4000	2 to 42	51	6x6	41	.031 to 1.3	2000	26000

NOTE: All requirements shall be not less than the figures shown. Where a range is shown, the requirements shall range from not more than the minimum to not less than the maximum figures shown in the tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.4.3.4 Tilt mechanism. The tilt portion of the class A, B and C positioner arm shall be supported by sealed, anti-friction, taper roller bearings. The tilt arm of each positioner shall be provided with an indexing pin locking system for locking the tilt mechanism in not less than eight equally spaced positions. In addition to the pin locking system, the tilt arm of each class A positioner shall be provided with a hand operated, adjustable, friction brake system. The powered tilt mechanism of class B and C shall be controlled by a forward and reverse toggle switch with a light to indicate when power is on.

3.4.3.5 Angle adjustment system. The arm angle adjustment for the class A, sizes 1a through 7 positioners, shall be accomplished through the use of a self-locking worm and wormgear sector system. The system shall be manually adjustable to any desired angle with a capacity load under all operating conditions.

3.4.4 Electrical requirements. The positioner shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each positioner shall be designed to operate on 3-phase, 60 Hertz (Hz), 230/460 volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The positioner shall be initially wired as specified (see 6.2).

3.4.4.1 Motors. Motors shall meet the requirements of 3.4.4. Motors shall be rated for heavy duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.4.2 Weld current grounding device. A weld current grounding device shall be furnished with each positioner. The capacity of the grounding device shall not be less than the amperes specified in manufacturer's commercial operator's manual. The positioner shall be provided with direct current (DC) with suitable frame to ground cable attachment. The attachment for securing the welding ground cable attachment to the positioner shall be permanently mounted in a position which will prevent interference during full operational ranges of the positioner. The type of connection shall ensure low resistance electrical conductivity. The weld current grounding device shall meet the requirements of NFPA 79.

3.4.5 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the positioners shall meet the requirements of tables I, II and III for the positioner class and size as specified. Positioner size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.6 Physical size limitations. When specified (see 6.2.), the physical size of the positioner shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.7 Performance. The positioner shall be capable of supporting, rotating, and tilting capacity loads as specified, and shall conform to the requirements of tables I, II, and III and additional requirements of this specification. In addition, size 4 through 7 of class A and B, and sizes 5 through 7 of class C positioners shall be capable of balancing a weldment when the center of gravity of the weldment has been located directly on the rotation axis of the work table by providing necessary table tilt and arm angle adjustments so that the rotation axis will intersect the center of gravity of the weldment.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial positioner shall be furnished.

3.9 Optional and accessory equipment. Optional and accessory equipment specified in the following subparagraph shall be furnished as specified (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or positioner.

a. Air or hydraulic tilt and table brake with foot control valve.
b. Extra work tables.
c. Three jaw self-cleaning chuck.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on the positioner including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.11.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each positioner. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.11.2 Nameplate. A nameplate shall be securely attached to each positioner. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Table RPM (when applicable)
Load capacity (pounds)
Tilt degrees (when applicable)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.12 Workmanship. Workmanship of the positioner and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.1.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The positioner shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each positioner prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.4. The positioner shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each positioner to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the positioner is installed at its final location. The acceptance test shall consist of the examination in 4.7 And all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each positioner shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each positioner shall be operated through all movements for a period of 30 minutes without a table load. The control program shall be cycled each axis through its entire range, addressing all speeds, feeds, direction of the table rotation, tilt action, and standard control functions specified in this specification. Angle adjustments shall be smooth and shall not show evidence of any galling or binding. Table tilt brake system and pin locking system shall be checked for proper operations. Motorized table positioners shall be connected to an electrical power source and proper of all positioner functions, controls, motors, adjusting mechanisms, and accessories shall be verified during the test period. The torque range accuracy shall be checked for compliance with the requirements specified herein.

4.8.3 Performance test. The positioner shall be provided with a capacity table load, in accordance with tables I, II, and III with the load secured to the table surface. All tests cited in 4.8.1 through 4.8.3 shall be repeated. With the table positioned over the base, floor mounted positioners shall be adjusted to the maximum vertical height and without the use of counterweights, the table shall be operated through all rotation, tilt, and angle movements without tilting at the base. DC amperage equal to not less than the ground

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

weld current amperage specified in commercial operator's manual for the size positioner ordered shall pass through the positioner with the table rotating for a period of at least one (1) minute with no current interruption or damage to the positioner. All performance tests shall be completed without making any adjustments to or performing any maintenance on the positioner.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 through 4.8.3.2 shall be as specified and fully described (see 6.2).

4.8.3.2 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the positioner offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided under the provisions of the contract.

4.8.3.3 Noise level test. Noise level measurements test shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements test shall be made at the operators work station and at each corner of the positioner work table. Noise emitted by the positioner when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type II sound level meter at slow response. The test shall be conducted with the positioner operating at intermediate and maximum speeds.

4.8.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8.5 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. Class A and B, size 1 positioner is intended primarily for use in controlled atmosphere chambers to position and rotate workpiece weldments for extremely precise welding processes. Class A and B sizes, 1a through 7, and class C, sizes 5 through 7, positioners are intended for use in production of welding processes to position, balance, rotate, and tilt workpiece weldment for downhand welding operation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

h. Noise level requirements,if different (see 3.2.5.2).

i. Specify environmental protection requirements exceeding EPA and OSHA requirements, if different (see 3.2.6).

j. Specify surface finish requirements, if different (see 3.3.3).

k. Painting, if different (see 3.3.5.).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Foot operated locks, if required (see 3.4.1.1).

n. Table configuration, if different (see 3.4.3).

o. Table work surface, if different (see 3.4.3).

p. Remote control for size 1 positioner, if required (see 3.4.3.2).

q. Voltage requirements, if different (see 3.4.4).

r. Initial voltage, as specified (see 3.4.4).

s. Motors, if different (see 3.4.4.1).

t. Worklight, if required (see 3.4.5).

u. Size and capacity, if different (see 3.5).

v. Specify physical size limitation, if required (see 3.5.1).

w. Additional performance features, if required (see 3.6.1.

x. Specify and fully describe optional equipment, if required (see 3.8).

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

y. Specify applicable repair parts, if required (see 3.9).

z. Lubrication plate, if required (see 3.10.1).

aa. Lubrication plate information, if different (see 3.10.1).

bb. Nameplate, if different (see 3.10.2).

cc. Inspection conditions, if different (see 4.3).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (4.5).

ff. Acceptance test, if different (see 4.6).

gg. Additional performance tests, if required (see 4.8.3.1).

hh. Noise level measurement test, if different (see 4.8.3.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-P-80186A	MIL-P-80186B
Class 1	Class A
Class II	Class B
Class III	Class C
	Size 6a (new size)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual positioner in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new positioner into its future environment.

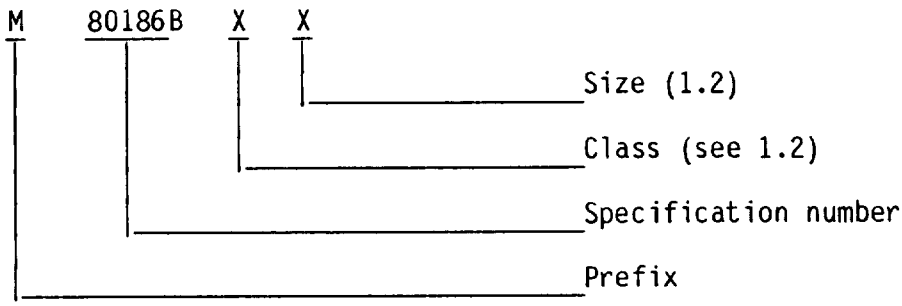
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for positioner acquired to this specification are created as follows:



6.14 Subject term Key word listing.

Angle adjustable system
Bench mounted positioner
Controlled atmosphere operated table.
Floor mounted positioner
Motorized table rotation
Positioner classes
Positioner tilt mechanism
Roller supported positioner
Steel swivel casters

Custodians:
Army - AR
Navy - SH
Air Force- 99

Review activities:
Army - AL
Air Force - 84
DLA - GS

User activities:
Army - SM
Navy - MC, OS

Preparing Activity

DLA - IP

Project 3436-0044

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-80196B	91 Mar 20

3. DOCUMENT TITLE: POSITIONERS, WELDING, ROTATING TABLE, CANTILEVER ARM DESIGN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901) 775-6699	(2) AUTOVON 682-6699
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.